

MD. MIFTAHUR RAHMAN SWAPNIL

Product Designer & Builder | AI, Frontend & Automation

Mirpur, Dhaka, Bangladesh

+880 1318-090383

linkedin.com/in/mr-swapnil

miftahurr503@gmail.com

mrswapnil.me

github.com/Swapnil-360



SUMMARY

Product Designer & Builder with an innovative and curious mindset, always exploring new ideas, technologies, and ways to turn them into working products. I enjoy building and experimenting with new things, whether it is to solve a real-world problem, learn something new, or gain hands-on experience. I use AI as a practical part of my workflow to simplify tasks, improve user experiences, and reduce time spent on repetitive work. Currently, I'm especially interested in AI automation and building smarter workflows that make everyday processes easier and more efficient.

CORE TECHNICAL COMPETENCIES

- **Product & UI/UX Design:** User Research, Wireframing, High-Fidelity Prototyping (Figma), Design Systems, Responsive Layouts, User Flow Mapping
- **Frontend Engineering:** Next.js 14/15 (App Router), React 18/19, TypeScript, Tailwind CSS, JavaScript (ES6+), HTML5/CSS3, Mobile-First Web
- **Backend & Databases:** Supabase (PostgreSQL, Row Level Security, Auth), Node.js, REST APIs, Google Drive API, Firebase Realtime Database
- **AI & Intelligent Systems:** Knowledge Graphs & RAG Architecture, Multi-LLM Orchestration (Gemini 2.5, OpenRouter), Relational GCN, Neo4j, PyTorch, Deep Learning for EEG
- **Tools & Automation:** Git & GitHub, n8n Automation, Telegram Bot API, Vercel, Docker (Basic), VS Code, LaTeX, Overleaf

FEATURED PROJECTS

Personal AI Assistant — Mikasa — Node.js, Telegram Bot API, Supabase, n8n, Multi-LLM Orchestration, Local PC Bridge **Active**

- Architected an autonomous personal AI operating system with persistent memory extraction, continuous learning, and multi-model failover (**Gemini 2.5 Flash & OpenRouter**).
- Engineered bidirectional Telegram bot controls, cross-platform cron automation, and secure local PC execution bridges for terminal and desktop operations.
- Integrated real-time database grounding with Supabase, storing structured memory categories and user preferences with conflict resolution.

Edu51Portal — Next.js 14, TypeScript, Supabase PostgreSQL, Google Drive API, Tailwind CSS **Live**

- Designed and launched an all-in-one academic platform serving **~100 active BUBT students**, delivering instant routine lookups, exam notices, and course materials.
- Integrated Google Drive API with Supabase to distribute 1,000+ academic PDFs securely with **\$0 cloud storage infrastructure overhead**.
- Engineered section-specific schedule filtering, fuzzy course search, and mobile-first responsive interfaces with sub-second page performance.

MuteBD — Flutter, Dart, Google Maps API, Background Geofencing, Firebase **Completed**

- Engineered a location-based mobile application that automatically silences smartphones in designated quiet zones (mosques, hospitals, libraries, classrooms) using GPS geofencing.
- Implemented battery-optimized background geofence listeners with customizable boundary radii and offline-first quiet zone caching.

PawfectMatch — React Native, Node.js, Firebase Realtime DB, AI Pet Care Assistant **Completed**

- Designed and developed a pet adoption mobile platform connecting verified rescuers with adopters, featuring AI-guided pet care recommendations and matchmaking.

ADDITIONAL FEATURED PROJECTS

OpusGen AI — Next.js, TypeScript, Generative Diffusion Models, REST APIs, Vercel

Live

- Built an AI creative marketing suite providing automated studio lighting, background replacement, and 4x neural upscaling for e-commerce merchant catalogs.
- Constructed batch processing export pipelines and multi-ratio ad canvas expansion, accelerating creative asset production from hours to under 30 seconds.

RESEARCH & PUBLICATIONS

Relation-Aware Graph Retrieval over a Curriculum Knowledge Graph for Prerequisite QA

Accepted (IEEE OMLET 2026)

- IEEE OMLET 2026 (Nairobi, Kenya) — Accepted with Minor Revision (Paper ID: 1017, Scheduled for IEEE Xplore & Scopus).
- Authors: Md. Jahidul Kamal Islam, **Md. Miftahur Rahman Swapnil**, Md. Asif Ali, Shrabani Das, Shefayatuj Johara Chowdhury (BUBT CSE).
- Engineered a curriculum knowledge graph RAG system with a 2-layer R-GCN encoder and DistMult decoder over **418 nodes and 558 typed edges** verified in Neo4j.
- Achieved **45.5% exact-set match** (vs. 22.7% text-RAG), 63.8% Entity F1, and a **100% correct abstention rate** on unanswerable questions with 0 query-time LLM overhead.

AI-Enabled Smart Classroom Monitoring and Safety Automation

Completed (BUBT CSE)

- Authors: Nishat Anjum Sara, Sheikh Shamia Hasan Nila, Md. Asif Ali, Md. Jahidul Kamal Islam, **Md. Miftahur Rahman Swapnil** (Supervised by Sadah Anjum Shanto, Assistant Professor).
- Engineered an IoT smart classroom safety automation system integrating sensor telemetry (DHT11, PIR, Ultrasonic, Sound) with automated door actuators and alert buzzers.
- Implemented dual operational presets (Normal Mode & Lecture Mode) integrated with Firebase Realtime Database and mobile dashboard telemetry.

EEG-Based Motor Imagery Classification

Ongoing / Running (2026)

- Investigating deep learning architectures for assistive neuro-rehabilitation and paralysis motor control interfaces.

EDUCATION

Bachelor of Science in Computer Science & Engineering (CSE)

CGPA: 3.60 / 4.00

Bangladesh University of Business and Technology (BUBT) | Intake 51

2022 – 2026 (Expected)

Higher Secondary Certificate (HSC) — Science

GPA: 5.00 / 5.00

Shaheed Police Smrity College

2021

Secondary School Certificate (SSC) — Science

GPA: 5.00 / 5.00

Kadirabad BL High School, Pirganj, Rangpur

2019

LEADERSHIP & COMMUNITY ENGAGEMENT

BASIS Students' Forum — BUBT Chapter — Member, Graphics Designer & Media and Publication Secretary

2023 – 2026

- Directed creative visual branding, digital promotional media, and publication outreach for tech seminars, workshops, and industry networking events.

BUBT IT Club — General Member

2022 – Present

- Active contributor to peer programming circles, tech workshops, and student developer initiatives.

Internal University Event Management — Active Coordinator & Event Organizer

2022 – Present

- Coordinated stage management, guest logistics, and event operations for university tech fests and academic seminars.

LANGUAGES

• Bangla — Native / First Language

• English — Professional Working Proficiency